

Introduction to the power flow analysis

ELEC0447 - Analysis of electric power and energy systems

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Overview

1. Introduction
2. The power flow equations
3. Power flow solution method
4. Slack Bus Selection and Distributed Slack
(Advanced section)

Introduction

The background of the slide features a minimalist design with teal-colored geometric shapes. Two large teal triangles point upwards from the bottom corners, meeting at a central point. Below this meeting point, a smaller, darker teal triangle points downwards. The remaining space at the top of the slide is white, providing a clean backdrop for the title text.

What is a Power Flow Analysis?

Power flow (or load flow) analysis determines the steady-state *electrical operating state* of an AC power system under specified generation, consumption, and voltage control conditions.

The State Variables to Determine:

- ▶ Voltage magnitude V_k at every bus k
- ▶ Voltage phase angle θ_k at every bus k

$$\bar{\mathbf{V}} = \begin{bmatrix} V_1 e^{j\theta_1} & \dots & V_n e^{j\theta_n} \end{bmatrix}^T$$

Derived Quantities (from $\bar{\mathbf{V}}$):

- ▶ Branch currents $\bar{I}_{km}$ and loadings
- ▶ Active/reactive flows P_{km}, Q_{km}
- ▶ Transmission losses $P_{\text{loss}}, Q_{\text{loss}}$
- ▶ Generator reactive outputs Q_{Gk}

The Core Principle: Once the complex bus voltages $\bar{\mathbf{V}}$ are known, every other physical quantity in the network is directly computed via Ohm's and Kirchhoff's laws.

Why Isn't This Standard Linear Circuit Analysis? I

Linear Circuit Theory:

- ▶ Independent current or voltage sources
- ▶ Nodal analysis with linear impedances:

$$\bar{\mathbf{I}} = \mathbf{Y}_{\text{bus}} \bar{\mathbf{V}} \implies \bar{\mathbf{V}} = \mathbf{Y}_{\text{bus}}^{-1} \bar{\mathbf{I}}$$

- ▶ Solved in a single linear step

Power System Reality:

- ▶ Loads consume constant **power** (P_L, Q_L) , largely independent of voltage
- ▶ Generators regulate active power P_G and voltage magnitude V_G
- ▶ Injections: $S_k = \bar{V}_k \bar{I}_k^* = \bar{V}_k (\mathbf{Y}_{\text{bus}} \bar{\mathbf{V}})_k^*$

Why Isn't This Standard Linear Circuit Analysis? II

The Non-Linearity Challenge

The relationship between power injections and bus voltages involves quadratic terms (V_k^2) and trigonometric functions of angle differences. This forms a system of **non-linear algebraic equations** that must be solved using iterative numerical algorithms!

Why Do We Use Power Flow? (Operations & Reliability)

Power flow is the foundational simulation tool for grid operators and utilities:

- ▶ **Operational Security & Real-Time Monitoring:**

- ▶ Verify that all transmission line and transformer currents stay within thermal limits ($I_{km} \leq I_{\max}$) to prevent line sag or cable damage
- ▶ Ensure bus voltages remain strictly within regulatory safety bounds ($0.95 \leq V_k \leq 1.05$ pu) to prevent equipment tripping

- ▶ **Contingency Analysis ($N - 1$ Security Criterion):**

- ▶ Power systems are legally required to withstand the sudden tripping of any single asset (line, transformer, generator) without causing cascading overloads
- ▶ TSOs (e.g. Elia, RTE, TenneT) run thousands of automated power flows every hour to screen for $N - 1$ vulnerabilities

Why Do We Use Power Flow? (Planning, Markets & Stability)

► Grid Expansion & Long-Term Planning:

- Forecast where bottlenecks will appear 5–20 years ahead
- Size and place new transmission lines, subsea cables, transformers, or reactive compensation devices (STATCOM, synchronous condensers)

► Renewables Integration & Electricity Markets:

- Calculate grid hosting capacity for massive offshore wind and utility-scale solar
- Verify the physical feasibility of market clearing schedules and compute Locational Marginal Prices (LMP) and transmission congestion charges

► Foundation for Dynamic Stability Studies:

- Power flow provides the initial equilibrium point ($\mathbf{V}_0, \boldsymbol{\theta}_0$) required to initialize transient stability, small-signal stability, and voltage stability simulations

Power Flow Problem Statement & Bus Types I

At every bus k , four physical quantities describe the state:

$$(V_k, \theta_k, P_k, Q_k)$$

In power flow analysis, **two variables are specified** (boundary conditions) and **two variables are unknown**:

- ▶ **PQ Buses (Load buses):**

- ▶ Active power P_k^{spec} and reactive power Q_k^{spec} are specified (consumed by loads, or injected by non-voltage-regulating renewable generators).
- ▶ **Unknowns to find:** Voltage magnitude V_k and phase angle θ_k .

Power Flow Problem Statement & Bus Types II

► PV Buses (Voltage-regulated generator buses):

- Active power P_k^{spec} and voltage magnitude V_k^{spec} are specified (controlled via governor and Automatic Voltage Regulator (AVR)).
- **Unknowns to find:** Phase angle θ_k and reactive power Q_k .
- *Reactive limits:* If Q_k hits a capability limit ($Q_{\min}$ or $Q_{\max}$), the generator cannot maintain voltage; the bus switches to a **PQ bus** with $Q_k = Q_{\text{limit}}$.

► One Slack Bus (Swing or Reference bus):

- Sets the angular reference for the entire system: $\theta_{\text{slack}} = 0^\circ$.
- Voltage magnitude is specified (typically $V_{\text{slack}} = 1.0$ pu).
- **Unknowns to find:** Active power P_{slack} and reactive power Q_{slack} .

Power Flow Problem Statement & Bus Types III

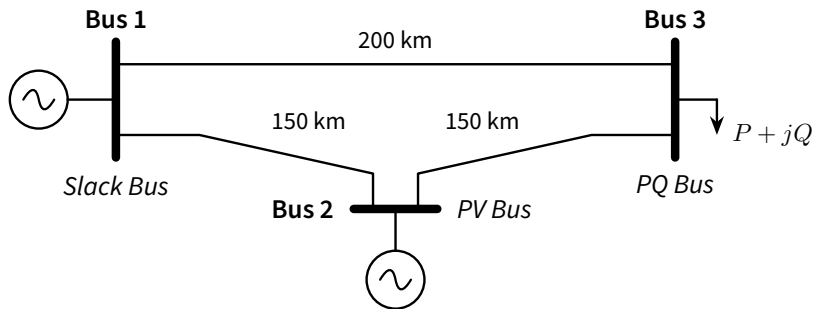
Why is a Slack Bus strictly necessary?

Transmission losses ($P_{\text{loss}} = \sum R_{km} I_{km}^2$) depend on the branch currents, which are unknown before voltages are found! Therefore, total generation cannot be fixed in advance:

$$\sum P_G = \sum P_L + P_{\text{loss}}(\bar{\mathbf{V}})$$

The slack bus generator automatically acts as the buffer that balances the remaining active and reactive power of the grid.

A First Tiny Example (3-Bus System) I



A First Tiny Example (3-Bus System) II

Buses:

- ▶ **Bus 1 (Slack):** $V_1 = 1.00$ pu, $\theta_1 = 0^\circ$
- ▶ **Bus 2 (PV):** $V_2 = 1.05$ pu, generating $P_2 = 2.0$ pu (200 MW)
- ▶ **Bus 3 (PQ):** Load consuming $P_3 = 5.0$ pu (500 MW) and $Q_3 = 1.0$ pu (100 Mvar)

Lines (Base: 345 kV, 100 MVA):

- ▶ $X = 0.376 \Omega/\text{km}$, $R = 0.037 \Omega/\text{km}$
- ▶ Lengths: $L_{12} = 150$ km, $L_{23} = 150$ km, $L_{31} = 200$ km
- ▶ Base impedance:
 $Z_B = 345^2/100 = 1190.25 \Omega$
- ▶ Current rating: $I_{\max} = 0.20$ kA

Modeling the System in PandaPower

We will see how to formulate mathematically the problem and solve it from scratch.

Usually, engineers use specialized power system software. One such open-source, Python-based tool is **PandaPower** [1].

► [Open PandaPower 3-Bus Model in Google Colab](#)

Creation of the Power Flow Model with PandaPower I

```
1  # Import PandaPower package and name it pp
2  import pandapower as pp
3
4  # Create an empty network object
5  net = pp.create_empty_network()
6
7  # Per unit bases (3-phase values)
8  Pbase = 100  # MVA
9  Vbase = 345 # kV
10
11 # Create buses (geodata are coordinates for graphical representation)
12 b1 = pp.create_bus(net, vn_kv=Vbase, name="Bus 1", geodata=(0,1))
13 b2 = pp.create_bus(net, vn_kv=Vbase, name="Bus 2", geodata=(2.5,0))
14 b3 = pp.create_bus(net, vn_kv=Vbase, name="Bus 3", geodata=(5,1))
```

Creation of the Power Flow Model with PandaPower II

```
15
16 # Create bus elements
17 pp.create_ext_grid(net, bus=b1, vm_pu=1.00, name="Grid Connection")
18 pp.create_load(net, bus=b3, p_mw=5*Pbase, q_mvar=1*Pbase, name="Load")
19 pp.create_gen(net, bus=b2, p_mw=2*Pbase, vm_pu=1.05, name="PV")
20
21 # Create branch elements.
22 # Here I neglect shunt capacitances.
23 Zbase = Vbase**2 / Pbase
24 X_km = 0.376
25 R_km = 0.037
26 l12_km = 150
27 l23_km = 150
28 l31_km = 200
```

Creation of the Power Flow Model with PandaPower III

```
29
30 pp.create_line_from_parameters(net, name="line1", from_bus = b1, to_bus = b2,
31     length_km=l12_km, r_ohm_per_km = R_km, x_ohm_per_km = X_km,
32     c_nf_per_km = 0, max_i_ka = 0.2)
33 pp.create_line_from_parameters(net, name="line2", from_bus = b2, to_bus = b3,
34     length_km=l23_km, r_ohm_per_km = R_km, x_ohm_per_km = X_km,
35     c_nf_per_km = 0, max_i_ka = 0.2)
36 pp.create_line_from_parameters(net, name="line3", from_bus = b3, to_bus = b1,
37     length_km=l31_km, r_ohm_per_km = R_km, x_ohm_per_km = X_km,
38     c_nf_per_km = 0, max_i_ka = 0.2)
39
40 # Solve the model
41 pp.runpp(net)
```

Results of the 3-Bus Example: Bus State

Bus	Type	V [pu]	θ [°]	P [MW]	Q [Mvar]
Bus 1	Slack	1.000	0.00	-308.38	+81.61
Bus 2	PV	1.050	-2.07	-200.00	-266.74
Bus 3	PQ	0.978	-8.79	+500.00	+100.00

Legend & Conventions:

- ▶ Standard text: Specified input data
- ▶ **Bold blue text: Computed by power flow**
- ▶ **Sign convention (PandaPower net flow):**
 - ▶ Negative (−): Net injection (generation)
 - ▶ Positive (+): Net withdrawal (load)

Physical Observations:

- ▶ Bus 1 generator supplies 308.38 MW to balance total load plus line losses
- ▶ Bus 2 generator injects 266.74 Mvar to hold voltage at 1.05 pu
- ▶ Voltage at load Bus 3 drops to 0.978 pu ($\Delta\theta = -8.79^\circ$)

Results of the 3-Bus Example: Line Flows & Loading

Line	From → To	P_{from} [MW]	P_{to} [MW]	P_{loss} [MW]	Q_{loss} [Mvar]	I [kA]	I_{max} [kA]	Loading [%]
Line 1	Bus 1 → Bus 2	+69.0	-68.2	0.80	8.08	0.22	0.20	109.8 %
Line 2	Bus 2 → Bus 3	+268.2	-264.2	3.97	40.30	0.49	0.20	245.2 %
Line 3	Bus 3 → Bus 1	-235.8	+239.4	3.62	36.75	0.40	0.20	201.2 %
Total Losses				8.39	85.13			

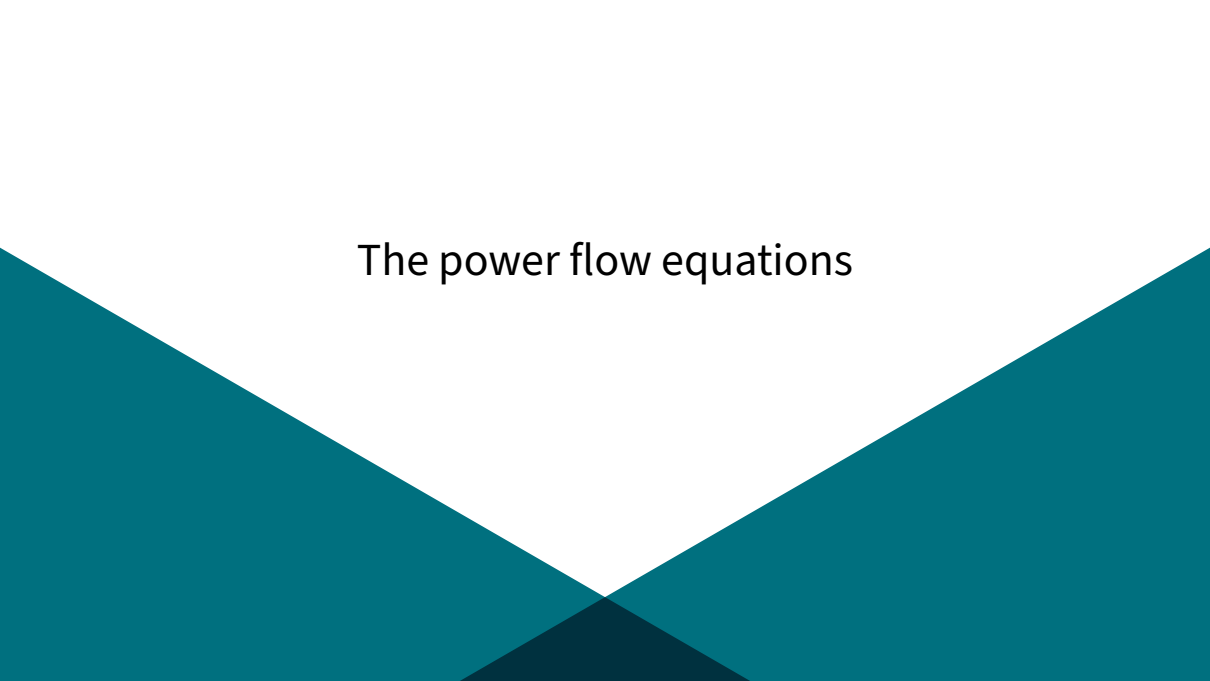
Active Power Balance:

- ▶ Total Gen: $P_G = 308.38 + 200.00 = 508.38$ MW
- ▶ Total Load: $P_L = 500.00$ MW
- ▶ Losses: $P_G - P_L = 8.38$ MW (1.65% of load)
- ▶ Matches sum of line losses: 8.39 MW

Engineering Diagnosis:

- ▶ Bus voltages are healthy (0.978 – 1.05 pu)
- ▶ **Lines 2 and 3 are severely overloaded** (245% and 201%)!
- ▶ Protection relays would trip these lines, risking cascading failure without re-dispatch!

The power flow equations

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The Nodal Admittance Matrix $\mathbf{Y}_{\text{bus}}$

Consider a power system with bus set $\mathcal{N} = \{1, \dots, n\}$ interconnected by transmission lines modeled by equivalent π circuits:

- ▶ For line connecting buses k and m : series impedance $\bar{z}_{km} = r_{km} + jx_{km}$, series admittance $\bar{y}_{km} = 1/\bar{z}_{km}$ (and $\bar{y}_{km} = 0$ if no line exists)
- ▶ Line total charging shunt susceptance b_{km}^{sh} , split equally as $j\omega C_{km}/2$ at each terminal
- ▶ Shunt element connected at node k (e.g. reactor/capacitor): $Y_{k,sh} = G_{k,sh} + jB_{k,sh}$
- ▶ Total shunt admittance between bus k and ground:

$$Y_{kG} = Y_{k,sh} + \sum_{m \in \mathcal{N}_k} j \frac{\omega C_{km}}{2}$$

The Nodal Admittance Matrix $\mathbf{Y}_{bus}$ II

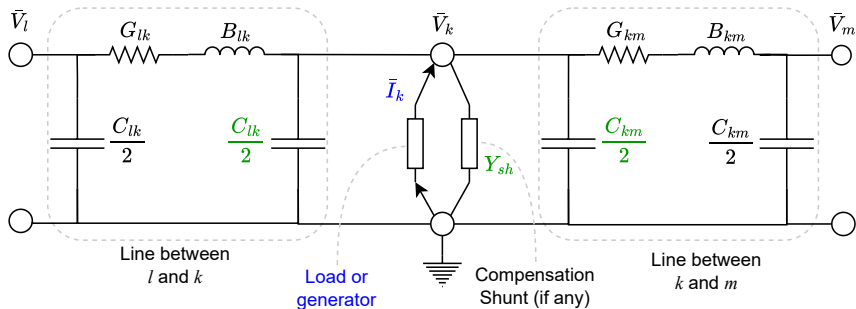


Figure 1: Current injection at bus k as a function of branch flows to neighbors (l, m) and shunt elements to ground.

The Nodal Admittance Matrix $\mathbf{Y}_{\text{bus}}$ III

By Kirchhoff's Current Law (KCL), the net current injection $\bar{I}_k$ into bus k is:

$$\bar{I}_k = Y_{kG} \bar{V}_k + \sum_{m \in \mathcal{N} \setminus \{k\}} \bar{y}_{km} (\bar{V}_k - \bar{V}_m) \quad (1)$$

Factoring out the bus voltages:

$$\bar{I}_k = \underbrace{\left(Y_{kG} + \sum_{m \in \mathcal{N} \setminus \{k\}} \bar{y}_{km} \right)}_{Y_{kk}} \bar{V}_k + \sum_{m \in \mathcal{N} \setminus \{k\}} \underbrace{(-\bar{y}_{km})}_{Y_{km}} \bar{V}_m \quad (2)$$

The Nodal Admittance Matrix $\mathbf{Y}_{\text{bus}}$ IV

In matrix form for all n buses:

$$\bar{\mathbf{I}} = \mathbf{Y}_{\text{bus}} \bar{\mathbf{V}} \quad (3)$$

Rules for building $\mathbf{Y}_{\text{bus}}$ by inspection:

- ▶ **Diagonal element Y_{kk} :** sum of all admittances connected to bus k (series lines + shunts)
- ▶ **Off-diagonal element Y_{km} ($m \neq k$):** negative of the series admittance connecting bus k and bus m (hence $Y_{km} = Y_{mk}$ for bilateral lines)

Power Flow Equations in Polar Coordinates I

In power systems, boundary conditions are given in powers rather than currents. The complex power injection vector is:

$$\mathbf{S} = \mathbf{P} + j\mathbf{Q} = \text{diag}(\bar{\mathbf{V}})\bar{\mathbf{I}}^* = \bar{\mathbf{V}} \circ (\mathbf{Y}_{\text{bus}}\bar{\mathbf{V}})^* \quad (4)$$

where $\circ$ denotes the Hadamard (element-wise) product.

For a single bus k :

$$S_k = P_k + jQ_k = \bar{V}_k \bar{I}_k^* = \bar{V}_k \sum_{m=1}^n Y_{km}^* \bar{V}_m^* \quad (5)$$

Power Flow Equations in Polar Coordinates II

Expressing voltages in polar coordinates and admittance in rectangular coordinates:

$$\bar{V}_k = V_k e^{j\theta_k}, \quad \bar{V}_m = V_m e^{j\theta_m}, \quad Y_{km} = G_{km} + jB_{km}$$

Then:

$$S_k = \sum_{m=1}^n V_k V_m e^{j(\theta_k - \theta_m)} (G_{km} - jB_{km})$$

Power Flow Equations in Polar Coordinates III

Using Euler's identity $e^{j\theta_{km}} = \cos \theta_{km} + j \sin \theta_{km}$ with phase difference $\theta_{km} = \theta_k - \theta_m$:

$$P_k = V_k \underbrace{\sum_{m=1}^n V_m (G_{km} \cos \theta_{km} + B_{km} \sin \theta_{km})}_{p_k(\mathbf{V}, \boldsymbol{\theta})} \quad (6)$$

$$Q_k = V_k \underbrace{\sum_{m=1}^n V_m (G_{km} \sin \theta_{km} - B_{km} \cos \theta_{km})}_{q_k(\mathbf{V}, \boldsymbol{\theta})} \quad (7)$$

Power Flow Equations in Polar Coordinates IV

Separating the self-term ($m = k$, where $\theta_{kk} = 0$, $\cos 0 = 1$, $\sin 0 = 0$):

$$P_k = G_{kk} V_k^2 + V_k \sum_{m \neq k} V_m (G_{km} \cos \theta_{km} + B_{km} \sin \theta_{km}) \quad (8)$$

$$Q_k = -B_{kk} V_k^2 + V_k \sum_{m \neq k} V_m (G_{km} \sin \theta_{km} - B_{km} \cos \theta_{km}) \quad (9)$$

- ▶ $G_{kk} V_k^2$ represents local active power dissipation
- ▶ $-B_{kk} V_k^2$ represents local reactive power injected by shunt susceptance
- ▶ The summation over $m \neq k$ represents the active/reactive power exchanged with neighboring buses

Number of Equations and Unknowns I

In an n -bus power system with n_{PQ} PQ buses, n_{PV} PV buses, and 1 slack bus ($n = n_{PQ} + n_{PV} + 1$):

Bus Type	Count	Specified Knowns	Iterative Unknowns	Equations Used
PQ Bus	n_{PQ}	$P_k^{\text{spec}}, Q_k^{\text{spec}}$	V_k, θ_k	$P_k - p_k = 0, Q_k - q_k = 0$
PV Bus	n_{PV}	$P_k^{\text{spec}}, V_k^{\text{spec}}$	θ_k	$P_k - p_k = 0$
Slack Bus	1	$V_s^{\text{spec}}, \theta_s^{\text{spec}} = 0$	none (in NR)	none (in NR)
Total	n	$2n$ knowns	$2n_{PQ} + n_{PV}$	$2n_{PQ} + n_{PV}$ equations

Number of Equations and Unknowns II

Post-processing quantities (computed after convergence):

- ▶ Slack bus power injections: $P_{\text{slack}} = p_{\text{slack}}(\mathbf{V}, \boldsymbol{\theta})$ and $Q_{\text{slack}} = q_{\text{slack}}(\mathbf{V}, \boldsymbol{\theta})$
- ▶ Generator reactive power at PV buses: $Q_k = q_k(\mathbf{V}, \boldsymbol{\theta})$ for $k \in \mathcal{N}_{PV}$
- ▶ Branch currents, active/reactive line flows, and system losses

Power flow solution method

The background of the slide features a white upper half and a teal lower half. The teal section is composed of two large triangles that meet at a point at the bottom center, creating a V-shape. The text "Power flow solution method" is centered in the white area.

Power Flow Solution via Newton-Raphson I

Let the specified (known) active and reactive powers be P_k^0 and Q_k^0 .

We define the **power mismatch functions**:

$$\Delta P_k(\mathbf{V}, \boldsymbol{\theta}) = P_k^0 - p_k(\mathbf{V}, \boldsymbol{\theta}), \quad \forall k \in \mathcal{N}_{PQ} \cup \mathcal{N}_{PV}$$

$$\Delta Q_k(\mathbf{V}, \boldsymbol{\theta}) = Q_k^0 - q_k(\mathbf{V}, \boldsymbol{\theta}), \quad \forall k \in \mathcal{N}_{PQ}$$

We want to find a solution such that

$$\Delta P_k(\mathbf{V}, \boldsymbol{\theta}) = \Delta Q_k(\mathbf{V}, \boldsymbol{\theta}) = 0,$$

i.e. we want to find values for unknown variables to explain the known variables.

Power Flow Solution via Newton-Raphson II

This forms a set of $2n_{PQ} + n_{PV}$ non-linear algebraic equations in $2n_{PQ} + n_{PV}$ unknown state variables:

$$\mathbf{x} = \begin{bmatrix} \boldsymbol{\theta}_{PQ,PV} \\ \mathbf{V}_{PQ} \end{bmatrix}, \quad \mathbf{F}(\mathbf{x}) = \begin{bmatrix} \Delta \mathbf{P}(\mathbf{x}) \\ \Delta \mathbf{Q}(\mathbf{x}) \end{bmatrix} = \mathbf{0}$$

The standard method to solve this non-linear system is the **Newton-Raphson method**.

Newton-Raphson Method in 1D: Principles

Suppose we want to find x such that the mismatch $g(x) = c - f(x) = 0$.

- ▶ Start with an initial guess $x^{(0)}$ at iteration $i = 0$
- ▶ First-order Taylor approximation around $x^{(i)}$:

$$g(x^{(i)} + \Delta x) \approx g(x^{(i)}) + g'(x^{(i)})\Delta x = 0$$

- ▶ Solve for the update step: $\Delta x = -\frac{g(x^{(i)})}{g'(x^{(i)})} = \frac{c - f(x^{(i)})}{f'(x^{(i)})}$
- ▶ Update iterate: $x^{(i+1)} = x^{(i)} + \Delta x$ and repeat until $|g(x^{(i)})| \leq \epsilon$

Quadratic Convergence: If $x^{(0)}$ is near the root, $|x^{(i+1)} - x^*| \leq K|x^{(i)} - x^*|^2$. The number of accurate decimal digits roughly doubles at each iteration!

Newton-Raphson 1D: Step-by-Step Convergence

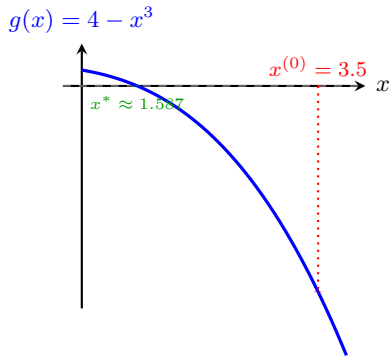
Example: Solve $g(x) = 4 - x^3 = 0$

$$\Delta x^{(i)} = \frac{4 - (x^{(i)})^3}{3(x^{(i)})^2}, \quad x^{(i+1)} = x^{(i)} + \Delta x^{(i)}$$

i	$x^{(i)}$	$g(x^{(i)})$	$\Delta x^{(i)}$	$ x^{(i)} - x^* $
0	3.5000	-38.88	-1.058	1.913
1	2.4422	-10.57	-0.590	0.855
2	1.8517	-2.35	-0.228	0.264
3	1.6233	-0.27	-0.035	0.036

► [Python Colab Implementation](#)

► [Interactive Web Visualizer](#)



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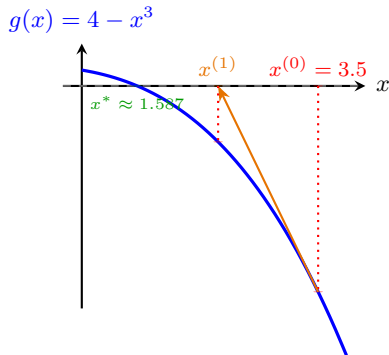
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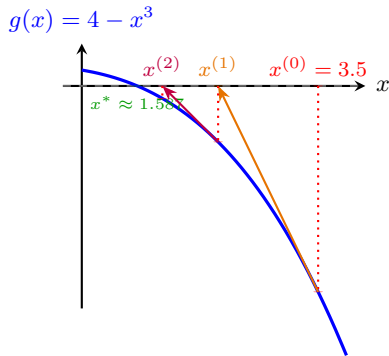
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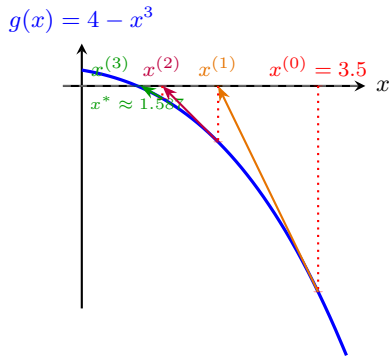
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Newton-Raphson for Power Flow I

We apply the same principle in dimension $M = 2n_{PQ} + n_{PV}$. At iteration (i) , the linearized mismatch system is:

$$\mathbf{J}(\mathbf{x}^{(i)})\Delta\mathbf{x}^{(i)} = \mathbf{F}(\mathbf{x}^{(i)}) \quad (10)$$

where the **Jacobian matrix** $\mathbf{J}$ contains the partial derivatives:

$$\begin{bmatrix} \mathbf{J}_{P\theta} & \mathbf{J}_{PV} \\ \mathbf{J}_{Q\theta} & \mathbf{J}_{QV} \end{bmatrix}^{(i)} \begin{bmatrix} \Delta\boldsymbol{\theta} \\ \Delta\mathbf{V} \end{bmatrix}^{(i)} = \begin{bmatrix} \mathbf{P}^0 - \mathbf{p}(\mathbf{V}^{(i)}, \boldsymbol{\theta}^{(i)}) \\ \mathbf{Q}^0 - \mathbf{q}(\mathbf{V}^{(i)}, \boldsymbol{\theta}^{(i)}) \end{bmatrix} \quad (11)$$

Newton-Raphson for Power Flow II

The four sub-matrices of the Jacobian are:

► $\mathbf{J}_{P\theta} = \frac{\partial \mathbf{P}}{\partial \boldsymbol{\theta}}$ of size $(n_{PQ} + n_{PV}) \times (n_{PQ} + n_{PV})$

► $\mathbf{J}_{PV} = \frac{\partial \mathbf{P}}{\partial \mathbf{V}}$ of size $(n_{PQ} + n_{PV}) \times n_{PQ}$

► $\mathbf{J}_{Q\theta} = \frac{\partial \mathbf{Q}}{\partial \boldsymbol{\theta}}$ of size $n_{PQ} \times (n_{PQ} + n_{PV})$

► $\mathbf{J}_{QV} = \frac{\partial \mathbf{Q}}{\partial \mathbf{V}}$ of size $n_{PQ} \times n_{PQ}$

Once $\Delta \boldsymbol{\theta}$ and $\Delta \mathbf{V}$ are calculated by solving the linear system:

$$\boldsymbol{\theta}^{(i+1)} = \boldsymbol{\theta}^{(i)} + \Delta \boldsymbol{\theta}, \quad \mathbf{V}^{(i+1)} = \mathbf{V}^{(i)} + \Delta \mathbf{V}$$

Remarks on Practical Implementation

- ▶ **Avoid Inversion:** We never compute the matrix inverse $\mathbf{J}^{-1}$; instead, we solve the linear system $\mathbf{J}\Delta\mathbf{x} = \Delta\mathbf{F}$ via LU factorization and forward/backward substitution
- ▶ **Sparsity Exploitation:** In real power systems, each bus is typically connected to only 2–5 neighboring lines. The Jacobian is extremely sparse (often $> 99\%$ zeros). Using sparse matrix algorithms reduces computational cost from $\mathcal{O}(n^3)$ to nearly $\mathcal{O}(n^{1.2})$!
- ▶ **Flat Start:** Without prior knowledge, iterations are initialized at a *flat start*:

$$V_k^{(0)} = 1.0 \text{ pu}, \quad \theta_k^{(0)} = 0^\circ \quad \forall k$$

- ▶ **Jacobian Reuse:** In modified Newton methods, the Jacobian can be held constant across multiple iterations close to convergence to save computation

Illustration on the 3-Bus Example I

Let us apply the Newton-Raphson algorithm step-by-step to our 3-bus network:

- ▶ Bus 1 is Slack ($V_1 = 1.0$ pu, $\theta_1 = 0^\circ$)
- ▶ Bus 2 is PV ($V_2 = 1.05$ pu, $P_2 = -2.0$ pu) $\rightarrow$ unknown: θ_2
- ▶ Bus 3 is PQ ($P_3 = 5.0$ pu, $Q_3 = 1.0$ pu) $\rightarrow$ unknowns: θ_3, V_3

The unknown vector has dimension 3:

$$\mathbf{x} = \begin{bmatrix} \theta_2 & \theta_3 & V_3 \end{bmatrix}^T$$

Illustration on the 3-Bus Example II

Step 0: Flat Start Initialization

$$\mathbf{x}^{(0)} = \begin{bmatrix} \theta_2^{(0)} & \theta_3^{(0)} & V_3^{(0)} \end{bmatrix}^T = \begin{bmatrix} 0 \text{ rad} & 0 \text{ rad} & 1.0 \text{ pu} \end{bmatrix}^T$$

Step 1: Initial Power Mismatches Evaluating p_2, p_3, q_3 at $\mathbf{x}^{(0)}$:

$$\begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \Delta Q_3 \end{bmatrix}^{(0)} = \begin{bmatrix} P_2^0 - p_2(\mathbf{x}^{(0)}) \\ P_3^0 - p_3(\mathbf{x}^{(0)}) \\ Q_3^0 - q_3(\mathbf{x}^{(0)}) \end{bmatrix} = \begin{bmatrix} -1.784 \\ +4.897 \\ -0.045 \end{bmatrix} \text{ pu}$$

Illustration on the 3-Bus Example III

Step 2: Linear System Solution Compute the 3×3 Jacobian at $\mathbf{x}^{(0)}$ and solve:

$$\begin{bmatrix} \frac{\partial p_2}{\partial \theta_2} & \frac{\partial p_2}{\partial \theta_3} & \frac{\partial p_2}{\partial V_3} \\ \frac{\partial p_3}{\partial \theta_2} & \frac{\partial p_3}{\partial \theta_3} & \frac{\partial p_3}{\partial V_3} \\ \frac{\partial q_3}{\partial \theta_2} & \frac{\partial q_3}{\partial \theta_3} & \frac{\partial q_3}{\partial V_3} \end{bmatrix}^{(0)} \begin{bmatrix} \Delta \theta_2 \\ \Delta \theta_3 \\ \Delta V_3 \end{bmatrix}^{(0)} = \begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \Delta Q_3 \end{bmatrix}^{(0)}$$

Update: $\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \Delta \mathbf{x}^{(0)}$.

Illustration on the 3-Bus Example IV

Convergence History of the 3-Bus System:

Iter i	$ \Delta P_2 $ [pu]	$ \Delta P_3 $ [pu]	$ \Delta Q_3 $ [pu]	Max Error
0	1.7840	4.8972	0.0451	4.8972
1	0.0206	0.1046	0.3125	0.3125
2	0.0005	0.0015	0.0033	0.0033
3	1.0×10^{-7}	2.4×10^{-7}	4.1×10^{-7}	4.1×10^{-7}

Convergence to 7 decimal places in only 3 iterations!

Final state: $\theta_2 = -2.066^\circ$, $\theta_3 = -8.787^\circ$, $V_3 = 0.9781$ pu, perfectly matching PandaPower!

► [Open 3-Bus Newton-Raphson Python Notebook](#)

Fast Decoupled Power Flow I

In high-voltage transmission networks, physical characteristics offer natural decoupling:

- ▶ Lines have high X/R ratios ($X \gg R$), so active power flow is predominantly coupled to voltage angles:

$$P \sim \frac{V_k V_m}{X} \sin(\theta_k - \theta_m) \quad \Rightarrow \quad \frac{\partial P}{\partial \theta} \gg \frac{\partial P}{\partial V}$$

- ▶ Reactive power flow is predominantly coupled to voltage magnitudes:

$$Q \sim \frac{V_k^2 - V_k V_m \cos \theta_{km}}{X} \quad \Rightarrow \quad \frac{\partial Q}{\partial V} \gg \frac{\partial Q}{\partial \theta}$$

Fast Decoupled Power Flow II

Setting the cross-coupling sub-matrices to zero ($\mathbf{J}_{PV} \approx \mathbf{0}$, $\mathbf{J}_{Q\theta} \approx \mathbf{0}$) splits the problem into two smaller, decoupled sub-systems:

$$\mathbf{J}_{P\theta}\Delta\theta = \Delta\mathbf{P} \quad (12)$$

$$\mathbf{J}_{QV}\Delta\mathbf{V} = \Delta\mathbf{Q} \quad (13)$$

- ▶ Faster iteration speed (two smaller linear systems instead of one large system)
- ▶ The sub-Jacobians can be approximated by constant matrices and inverted/factorized only once
- ▶ Sub-Jacobians also provide direct sensitivity information useful for operational planning

DC Power Flow (Linear Approximation) I

The "**Direct Current**" (DC) **power flow** is a linear approximation of the AC power flow, widely used in wholesale market clearing, unit commitment, and long-term transmission planning:

Underlying Assumptions:

1. **Lossless Lines:** $R \ll X$, resistances are neglected ($R = 0$)
2. **Flat Voltages:** $V_k \approx 1.0$ pu for all buses
3. **Small Phase Differences:** $\theta_{km} \ll 1$ rad $\implies \sin \theta_{km} \approx \theta_{km}, \cos \theta_{km} \approx 1$
4. Reactive power flows are neglected; shunt admittances are ignored

DC Power Flow (Linear Approximation) II

Under these assumptions, the branch flow from bus k to bus m simplifies to:

$$P_{km} = \frac{V_k V_m}{X_{km}} \sin(\theta_k - \theta_m) \approx \frac{\theta_k - \theta_m}{X_{km}} = B_{km}(\theta_k - \theta_m)$$

And the bus power injection equation becomes strictly **linear**:

$$P_k = \sum_{m \in \mathcal{N}_k} B_{km}(\theta_k - \theta_m)$$

In matrix form:

$$\mathbf{P} = \mathbf{B}'\boldsymbol{\theta} \tag{14}$$

where $\mathbf{B}' = \Im(\mathbf{Y}_{\text{bus}})$ (without slack bus row/column).

Application of DC Power Flow

Application to the 3-Bus Case

- ▶ Compare AC power flow versus DC linear approximation
- ▶ Quantify the impact of neglecting losses and voltage variations
- ▶ Understand the code and analyze the results:

▶ [Open 3-Bus DC Power Flow Python Notebook](#)

Slack Bus Selection and Distributed Slack (Advanced section)

The Dual Role of the Slack Bus

Reminder. In classical AC power flow, the **slack (or swing) bus** fulfills two distinct roles:

1. Mathematical: Angle reference

- ▶ In AC grids, power flows depend strictly on **phase differences** $\theta_k - \theta_m$.
- ▶ Shifting all angles by $\Delta\theta$ leaves physical flows unchanged.
- ▶ One bus must anchor the coordinate frame:

$$\theta_{\text{ref}} = 0^\circ$$

removing the singularity of the Jacobian.

2. Physical: Power Balancing

- ▶ Active power conservation requires:

$$\sum P_{Gk} = \sum P_{Lk} + P_{\text{loss}}(\mathbf{V}, \boldsymbol{\theta})$$

- ▶ Total transmission losses P_{loss} depend on the unknown voltages $\mathbf{V}$.
- ▶ P_{slack} must remain a free variable to absorb the net loss mismatch.

How to Choose the Slack Bus in Single-Slack Studies I

How should the engineer select the slack bus?

► 1. Large Generating Capacity and Headroom:

- Connect to a major, flexible generating plant (e.g., large hydro or CCGT).
- Must have ample active and reactive spinning reserves (see lecture on generators):

$$P_{G,\min} \ll P_{G,\text{slack}} \ll P_{G,\max}, \quad Q_{G,\min} \ll Q_{G,\text{slack}} \ll Q_{G,\max}$$

to absorb unexpected loss variations without breaching machine capability limits.

► 2. Voltage Regulation Capability:

- Strong Automatic Voltage Regulator (AVR) and high reactive capability (Q) to maintain terminal voltage $V_{\text{slack}} = V^0$ (see lecture on generators).

How to Choose the Slack Bus in Single-Slack Studies II

▶ 3. Electrical Centrality and Grid Strength:

- ▶ High short-circuit capacity (low Thévenin impedance, electrically "stiff" bus).
- ▶ Centrally located in terms of electrical distance (impedance) to major load centers.
- ▶ High nodal degree: interconnected with multiple strong, high-capacity transmission lines.

▶ 4. Grid Interconnection / Tie-Line Substation:

- ▶ In regional transmission or distribution utility models, the boundary substation connecting to the external transmission grid is selected as the slack ("infinite bus").

Pitfalls of a Poor Slack Bus Choice I

Designating a single, arbitrary slack bus can create serious modeling distortions:

1. Artificial Congestion and False Overloads

- ▶ If a small or peripheral generator is chosen as slack, it is forced to supply the loss balance for the *entire* grid.
- ▶ All loss-compensating energy must travel across local transmission paths, triggering **spurious overload alarms** on adjacent lines that do not exist in reality!

2. Generator Limit Violations & Solver Divergence

If the single slack generator hits its $P_{\max}$ or $Q_{\max}$ limits, the solver must dynamically switch the slack bus, frequently causing limit-cycling or non-convergence in NR.

Pitfalls of a Poor Slack Bus Choice II

3. Sensitivity and Market Distortions (notions defined in the Energy Markets course)

- ▶ Sensitivities such as Power Transfer Distribution Factors (PTDF) and Locational Marginal Prices (LMP) depend directly on the slack reference
- ▶ An arbitrary slack choice skews market signals and transmission loss allocation.

Why Single Slack Fails in Real Practice I

The Physical Reality of Power Systems:

- ▶ In actual real-time operations, **no single power plant** absorbs the entire system mismatch or covers all grid losses!
- ▶ We will come back to that in the lecture on frequency stability!

1. Primary Frequency Control (Governor Droop, $t < 10$ s):

- ▶ Any power imbalance creates an instantaneous frequency deviation $\Delta f = f - f_0$.

Why Single Slack Fails in Real Practice II

- ▶ All synchronized generators equipped with speed governors respond **simultaneously** according to their droop characteristic R_k :

$$\Delta P_{Gk} = -\frac{1}{R_k} \Delta f$$

- ▶ The deficit or surplus is automatically shared across **all** online generators!

2. Secondary Frequency Control / AGC ($t \sim 30 \text{ s} - 15 \text{ min}$):

- ▶ Automatic Generation Control (AGC) restores frequency and tie-line schedules by dispatching regulation reserves across designated generators according to participation factors α_k .

Why Single Slack Fails in Real Practice III

The $N - 1$ Contingency Paradox

- ▶ Suppose a 1000 MW nuclear unit trips unexpectedly.
- ▶ In a **single-slack** power flow, the solver forces the single slack bus to pick up all 1000 MW. The resulting simulated line flows and voltages around that bus are completely unphysical!
- ▶ In **reality**, the 1000 MW deficit is shared among hundreds of generators across the entire synchronous interconnection (e.g. continental Europe).

The Distributed Slack Formulation I

To mirror physical reality, modern tools use a **Distributed Slack (or Distributed Balancing)** model:

- ▶ Assign **participation factors** $\alpha_k \geq 0$ to a set of participating generators $\mathcal{G}_{\text{slack}}$:

$$\sum_{k \in \mathcal{G}_{\text{slack}}} \alpha_k = 1$$

- ▶ Introduce a global scalar balancing variable: ΔP_{slack}
- ▶ Active power generation at each participating bus k becomes:

$$P_{Gk} = P_{Gk}^0 + \alpha_k \Delta P_{\text{slack}}$$

The Distributed Slack Formulation II

- Maintain **one** bus as the **angle reference**: $\theta_{\text{ref}} = 0^\circ$ (decoupling angle reference from power balancing!)

Closing the Power Balance System:

- Add the global active power balance equation to the system:

$$\sum_{k=1}^n P_k(\mathbf{V}, \boldsymbol{\theta}) = \sum_{k \in \mathcal{G}} P_{Gk} - \sum_{k \in \mathcal{L}} P_{Lk} - P_{\text{loss}}(\mathbf{V}, \boldsymbol{\theta}) = 0$$

- Total unknowns: $M = (2n_{PQ} + n_{PV}) + 1$ (the standard unknowns plus ΔP_{slack})
- Total equations: $M = (2n_{PQ} + n_{PV}) + 1$
- The augmented Jacobian remains square and is solved directly via Newton-Raphson!

Choosing Participation Factors α_k I

Depending on the engineering application, participation factors α_k are chosen as:

1. Governor Droop Based (Primary Frequency Response):

$$\alpha_k = \frac{P_{Gk}^{\max} / R_k}{\sum_j P_{Gj}^{\max} / R_j}$$

Faithfully reproduces post-contingency steady-state flows for $N - 1$ security analysis.

Choosing Participation Factors α_k II

2. Generation Capacity Proportional:

$$\alpha_k = \frac{P_{Gk}^{\max}}{\sum_j P_{Gj}^{\max}} \quad \text{or} \quad \alpha_k = \frac{P_{Gk}^0}{\sum_j P_{Gj}^0}$$

Distributes losses proportionally across generators without overloading individual units.

Choosing Participation Factors α_k III

3. Load Proportional (Distributed Load Slack):

$$\alpha_k = \frac{P_{Lk}}{\sum_j P_{Lj}}$$

Allocates losses as incremental demand across consumers (standard in market clearing and loss allocation).

4. Economic Merit-Order Participation:

$$\alpha_k \propto \frac{1}{\text{Marginal Cost}_k}$$

Distributes loss compensation according to optimal redispatch / tertiary reserves.

Distributed Slack in Industry Practice I

Key Engineering Takeaway

Decouple the angle reference from loss balancing:

- ▶ **Angle reference** ($\theta_{\text{ref}} = 0^\circ$) is a *mathematical necessity* to anchor the coordinate frame.
- ▶ **Distributed slack** (α_k) is a *physical necessity* to reflect real frequency response and prevent localized flow distortion.

Distributed Slack in Industry Practice II

Industrial Implementation:

- ▶ Supported natively in power systems software:
 - ▶ **PandaPower**: `pp.runpp(net, distributed_slack=True)`
 - ▶ **MATPOWER**: User-defined generator participation in OPF/PF
 - ▶ **PSS/E & PowerFactory**: Inertial/governor dispatch modes for $N - 1$ contingency screening
- ▶ **Electricity Market Clearing**:
 - ▶ In wholesale electricity markets (e.g. European Day-Ahead Flow-Based Market Coupling, US ISOs), distributed slack ensures that PTDFs and LMPs are neutral and do not favor an arbitrary bus.

References I



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